



Alberta Public Health Disease Management Guidelines

Meningococcal Disease, Invasive



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Alberta Health

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Revisions

Revision Date	Document Section	Description of Revision
February 2025	General	Updated Template
	Public Health Management	Revised Post Exposure Prophylaxis (PEP) for Close Contacts to align with Canadian Immunization Guide.

Case Definition

Confirmed Case

Clinical evidence of invasive disease^(A) with laboratory confirmation of infection:

- Isolation of *Neisseria (N.) meningitidis* from a normally sterile site (blood, cerebrospinal fluid [CSF], synovial, pleural or pericardial fluid),

OR

- Detection of *N. meningitidis* deoxyribonucleic acid (DNA) from a normally sterile site (blood, cerebrospinal fluid [CSF], synovial, pleural or pericardial fluid) using a validated molecular assay (e.g., PCR).

Probable Case

Clinical evidence of invasive disease^(A) with purpura fulminans or petechiae and no other apparent cause and with non-confirmatory laboratory evidence:

- With detection of *N. meningitidis* antigen in the CSF,

OR

- In the absence of isolation of *N. meningitidis* from a normally sterile site or in the absence of demonstration of *N. meningitidis* nucleic acid from a normally sterile site.

NOTE: The following cases^(B) require public health follow-up as they may require prophylaxis of contacts. Refer to [Post-Exposure Prophylaxis for Contacts section](#) for more information.

Primary Meningococcal Conjunctivitis Case

- Isolation of *N. meningitidis* from the eye or the conjunctival sac in association with purulent conjunctivitis.

Meningococcal Pneumonia Case

- Clinical or radiological evidence of pneumonia with laboratory confirmation of infection:
 - Presence of gram-negative diplococci on a Gram stain and a polymorphonuclear cell response from sputum or respiratory aspirate,**AND**
 - Isolation with heavy growth^(C) of *N. meningitidis* from an appropriate respiratory specimen (e.g., sputum or respiratory aspirate).

^(A) Clinical evidence of invasive disease includes meningitis and/or septicemia, orbital cellulitis, septic arthritis, pericarditis and pneumonia. Invasive disease may progress rapidly to purpura fulminans, shock and death.

^(B) These cases are NOT reported to Alberta Health.

^(C) Heavy growth is defined as 2+.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Zone Medical Officer of Health (MOH) (or designate) of all confirmed and probable cases in the prescribed form by the Fastest Means Possible (FMP).

Laboratories

All laboratories shall report all positive laboratory results:

- by FMP to the Zone MOH (or designate), and
- by mail, fax or electronic submission (e.g., HL7 message) within 48 hours (two business days) to the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and Indigenous Services Canada

- The Zone MOH (or designate) where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed and probable cases.
- The Zone MOH (or designate) where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country cases, the following information should be forwarded to the CMOH (or designate) by FMP:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical / epidemiological information.
- For out-of-province and out-of-country susceptible contacts, the following information should be forwarded to the CMOH (or designate) as soon as possible:
 - name,
 - date of birth, and
 - out-of-province/country contact information.

Epidemiology

Etiology

N. meningitidis is a gram-negative diplococcus. Thirteen distinct serogroups (based on their capsular polysaccharide) have been identified. Serogroups A, B, C, Y, X and W (previously W-135) are responsible for most of the invasive meningococcal disease (IMD) worldwide.^(1–4)

Clinical Presentation

Asymptomatic carriage occurs in up to 10% of the general population and should not be confused with meningococcal infection.^(5,6) Initial symptoms of infection can be nonspecific and resemble self-limiting illnesses such as viral respiratory infections.⁽⁵⁾ Invasive meningococcal disease (IMD) most commonly presents as either meningitis, meningococemia (meningococcal sepsis) or both.^(2,7,8)

Meningitis

Meningitis occurs in approximately 50% of persons with invasive disease and is characterized by abrupt onset of fever, intense headache, chills, malaise, prostration, nausea, vomiting, nuchal rigidity, and photophobia.^(1,2,7,8) In infants and toddlers, symptoms may have a slower onset, signs may be subtle and non-specific such as poor appetite, lethargy and irritability.⁽⁹⁾ Neck stiffness may be absent and a bulging fontanelle may be seen.^(10,11)

Meningococemia (sepsis)

Approximately 35–40% of IMD cases present as meningococemia, which is the most severe form of invasive disease.⁽²⁾ Early symptoms of sepsis include cold hands and feet, leg pains, and pallorous or mottled skin color. Symptoms of meningococemia also include sudden onset of fever, a characteristic petechial or purpuric rash which may progress to purpura fulminans, circulatory collapse, and death can occur within hours of onset.^(2,11)

Other less common manifestations of meningococcal disease include otitis media, septic arthritis, pneumonia, pericarditis and epiglottitis.^(8,11) Even with antibiotic treatment, the case-fatality rate of meningococcal disease is 10–15%, but can be as high as 50% for untreated cases.⁽¹²⁾ Neurologic disabilities, vision abnormalities, seizures, skin scarring, hearing loss, limb or digit amputations are permanent sequelae experienced by up to 20% of those who survive infection.^(1,5,10,13)

Reservoir

Humans.

Diagnosis

The diagnosis of meningococcal disease is by isolation of *N. meningitidis* or detection of meningococcal DNA from a normally sterile site.⁽¹³⁾ The detection of gram-negative diplococci in a sterile site may be used to make a presumptive diagnosis. Detection of *N. meningitidis* antigen provides rapid results compared to culture or PCR, however the sensitivity and specificity of this test is low. In addition, the test does not allow determination of serogroup and is therefore considered non-confirmatory laboratory evidence of disease.⁽¹⁴⁾ Positive antigen test results from urine and serum samples are unreliable for diagnosing meningococcal disease.⁽¹⁵⁾

The Public Health Laboratories (ProvLab) performs serogroup identification for all *N. meningitidis* isolates collected from sterile sites. The specimen is also sent to the National Microbiology Laboratory (NML) for serotyping (different from serogrouping) and further bacteriologic studies. For more information on specimen collection, refer to [ProvLab's Guide to Services](#).

Treatment

- Antibiotic treatment should begin immediately after a presumptive diagnosis in the case.
- Recommended medications for treatment are outlined in [Appendix 1](#).
- Alberta Health will provide publicly funded medications for the treatment of confirmed and probable cases.
- If the case is not treated with ceftriaxone or cefotaxime, ensure rifampin is also given prior to hospital discharge to eradicate nasopharyngeal carriage of *N. meningitides*.^(2,16)

Transmission

The primary mode of transmission is by direct contact of respiratory droplets or oral secretions from the nose and throat of an infected person or carrier.^(1,17) Approximately 10% of the population are asymptomatic carriers.^(8,11)

Incubation Period

The incubation period varies from two to 10 days, most commonly three to four days.⁽⁸⁾

Period of Communicability

IMD is communicable from seven days prior to symptom onset until 24 hours following the initiation of appropriate antibiotic therapy.^(2,11)

Host Susceptibility

Risk factors for IMD include host, organism and environmental factors.⁽¹¹⁾ Refer to Table 1 for more information.

Table 1: Examples of Risk Factors^(1,7,11,13)

Risk Factor	Examples
Age	• Children < 2 years of age are at greatest risk • Followed by adolescents and young adults aged 16–21 years • Adults > 65 years
Close contact with a case	• Household members • Persons who share sleeping arrangements with the case • Daycare workers & attendees
Living in closed/crowded conditions	• Military groups • Post-secondary students living in dormitories
Immune suppressed	• HIV infection • Anatomic or functional asplenia
Travel to hyperendemic/ epidemic areas	• African meningitis belt • During the Hajj in Saudi Arabia
Other	• Smoking or exposure to second hand smoke • Recent respiratory viral infection

Incidence

N. meningitidis has been responsible for both sporadic cases and disease outbreaks globally. The disease occurs year-round but has a seasonal pattern with a peak incidence occurring during winter and spring. The most common serogroups reported in Canada are B, C, W and Y.^(6,17)

Prior to the introduction of the meningococcal immunization program, children under one year of age in Alberta suffered the highest rates of IMD. The total number of IMD cases has declined dramatically in Alberta since the introduction of meningococcal serogroup C vaccine in 2002 as part of the universal infant immunization program as well as the meningococcal quadrivalent conjugate vaccine in 2011, as part of the school immunization schedule.^(18,19) There was an increase in the number of IMD cases reported from 2017 to 2018. This increase was mainly driven by serogroup W. Canada and several other countries also reported an increase in serogroup W cases.⁽²⁰⁾ Annual case counts can be accessed through Alberta Health's [Interactive Health Data Application \(IHDA\)](#).

Public Health Management

Key Investigation

- Confirm diagnosis and that individual meets case definition.
- Obtain history of illness including the date of onset, signs and symptoms.
- Identify risk factors for acquiring invasive meningococcal disease including history of recent travel or exposure to a confirmed case.
- Determine the possible source of infection taking into consideration communicability, incubation period, and mode of transmission.
- Determine eligibility and immunization history specific to meningococcal disease:
 - number of doses,
 - date administered,
 - where the person was immunized (e.g., out of country),
 - type of immunization provider (e.g., public health, doctor's office, travel clinic), and
 - if not immunized, determine reason why.
- Identify **close contacts** (see Table 2) who may have had prolonged exposure to the case within the seven days prior to onset of symptoms in the case and up to 24 hours after the case commences appropriate antibiotic therapy.^(2,21)

Table 2: Definition of Close Contacts^(1,2,6)

Close Contacts
• Individuals living in the same household* as the case or sharing sleeping arrangements with the case. • Staff and children in child care facility or nursery school. • Individuals who have had direct contact with the oral/nasal secretions of the case (e.g., kissing, shared cigarettes, food, glasses/bottles, eating utensils). • Health care workers who have had intensive unprotected contact (without wearing a mask) with infected patients (i.e., intubating, resuscitating, or closely examining the oropharynx). • Airline passengers sitting immediately on either side of the case (but not across the aisle) when the total time spent aboard the aircraft was at least eight hours.

**NOTE: contacts who live in the same household as the index case are 500 to 1200 times more likely to develop IMD than the general population. The increased risk for disease for household contacts may extend for up to one year after the disease in the case, well beyond any protection derived from chemoprophylaxis.⁽¹⁾*

Management of a Case

- In addition to routine practices, hospitalized individuals should be placed under droplet precautions until 24 hours of appropriate antibiotic therapy have been completed.^(2,8)
- If eligible according to the current [Alberta Immunization Policy \(AIP\)](#), cases who are unimmunized or partially immunized should be offered meningococcal-containing vaccine.

Management of Contacts

- Determine type of exposure during the seven days before onset of illness in the case and 24 hours after the case initiated appropriate antibiotic therapy.
- Determine eligibility for post-exposure prophylaxis (PEP). Refer to the section [Post Exposure Prophylaxis for Contacts](#) for more information.
- Determine meningococcal immunization history (i.e., type of vaccine, number of doses and date of administration).
- Close contacts who are unimmunized or partially immunized should be offered an age-appropriate dose of meningococcal-containing vaccine if they are eligible according to the current [AIP](#).
- Provide information about meningococcal disease, including signs and symptoms.
- Refer symptomatic contacts for assessment as appropriate.
- Advise asymptomatic contacts to monitor closely for symptoms and to seek immediate medical assessment if they develop febrile illness or any other signs or symptoms of meningococcal infection within 14 days following their last exposure to the case.

Post Exposure Prophylaxis (PEP) for Contacts

- Regardless of immunization status, chemoprophylaxis should be offered as soon as possible preferably within 24 hours of case identification.
- Chemoprophylaxis should be offered to **close contacts** (See [Appendix 1](#)) for the following cases:^(16,21,22)
 - IMD,
 - primary meningococcal conjunctivitis (PMC), and
 - meningococcal pneumonia.
- See Appendix 1 for recommended medications for PEP.
- Chemoprophylaxis is unlikely to be of benefit if the last exposure has exceeded 14 days or more.⁽⁷⁾
- Certain close contacts may also be offered immunoprophylaxis if the serogroup identified in the case is vaccine preventable. Serogroup-specific immunoprophylaxis may reduce the risk of subsequent meningococcal disease.^(1,2)
- The following close contacts should receive chemoprophylaxis and immunoprophylaxis:
 - Individuals living in the same household as the case or sharing sleeping arrangements with a case.
 - Staff and children in child care facility or nursery school.
 - Individuals who have had direct contact with the oral/nasal secretions of the case (e.g., kissing, shared cigarettes, food, glasses/bottles, eating utensils).
- The following close contacts should receive chemoprophylaxis only, immunoprophylaxis is not necessary⁽¹⁾:
 - Health care workers who have had intensive unprotected contact (without wearing a mask) with infected patients (i.e., intubating, resuscitating, or closely examining the oropharynx);
 - Airline passengers sitting immediately on either side of the case (but not across the aisle) when the total time spent aboard the aircraft was at least 8 hours; or
 - Close contacts of a case when the serogroup in the index case has not been determined.
- Refer to the [AIP](#) for current immunization recommendations.

NOTE: There is NO evidence to support the provision of widespread chemoprophylaxis for persons who are casual contacts of sporadic cases (e.g., health care workers that do not have risk of ongoing exposure, school/classroom,

workplace, social interactions) unless there is direct contamination of the nose or mouth with oral and/or nasal secretions of the case.⁽²⁾

Preventative Measures

- Promote immunization for all infants, adolescents, and high-risk groups. Refer to the [AIP](#) for current immunization recommendations.
- Provide public information about the risks of disease transmission and the importance of good hand hygiene and respiratory etiquette.
- Recommend to travelers that they visit their health care provider for consultation prior to their travel as immunization may be recommended/required prior to travel to destinations where meningococcal infection is hyper-endemic.
- Advocate for reduction of overcrowding in living quarters and workplaces.

Appendix 1: Recommended Antibiotics for Treatment and PEP

Antibiotic	Dosage	Comments
Rifampin	Infants < 1 months: 5 mg/kg per dose orally every 12h for 2 days (4 doses) Children ≥ 1 month: Maximum of 10 mg/kg (maximum 600 mg) orally every 12h for 2 days (4 doses) Adults ≥ 18 years of age: 600 mg orally every 12h for 2 days (4 doses)	• Contraindicated in pregnancy and persons with liver disease • Interferes with oral contraceptives, some anticonvulsants and anticoagulants • Stains soft contact lenses
Ceftriaxone	<12 years of age: 125 mg IM in a single dose Children ≥ 12 years and adults: 250 mg IM in a single dose	• Alternative for: - pregnant women - persons with liver disease or - allergy to rifampin • Dilute in 1% lidocaine to reduce pain at injection site
Ciprofloxacin	Adults ≥18 years of age: 500mg orally in a single dose	• Alternative for persons allergic to rifampin or ceftriaxone or unable to give IM injection • Contraindicated in pregnancy and lactation • A single dose medication regimen may improve compliance in some populations • Safe in liver disease

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